DO NOW — RECALL

Data Structures · Classes and Objects in C#

Name: _____ Class: _____ Date: _____

1. What is a method parameter?

2. What is a breakpoint?

3. How does a queue differ from stack?

4. Evaluate (A AND NOT B) OR C when A=1, B=0 and C=0. Show the intermediate Boolean values.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Named input to a method

2. Model answer

A debugger stopping point

3. Model answer

FIFO versus LIFO

4. Model answer

NOT B=1; A AND NOT B=1; then 1 OR C=1. Final output: 1.

By the end, you can…

• Define a class and instantiate an object

• Explain fields and methods

• Construct a simple record-like object

Vocabulary — know these terms

Recall the term — class

Recall the term — object

Recall the term — field

Recall the term — method

Recall the term — instance

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Independent state

first.Value = 2 and second.Value = 0 . Both calls act on first ; they do not change second .

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Predict before checking

Which instance runs Increase() ? Which instance field changes?

Targeted update

first.Value = 2 ; second.Value = 1 . The variable second refers to the second created Counter instance.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: defining a class automatically creates an object.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

NEW LEARNING

Read, annotate and explain

Separate blueprint and instance

```
Student pupil = new Student();
pupil.Name = "Asha";

class Student {
public string Name = "";
}
```

Class ≠ instance

A class defines the fields and behaviour available to each object. new Student() constructs an instance; a separate new creates another independent object.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Separate blueprint and instance



Three-stage diagram showing retrieve, model, and practise

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Explain why two objects of the same class may hold different field values. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Write C# to define a class Book with public string Title. [3]

Question 2

Apply independently
Instantiate a Book object named novel. [2]

I DO — WORKED EXAMPLE

Follow the worked method

One class, two independent objects

Identify class, objects and members

Badge is the class (blueprint). amber and blue are references to two objects . Each has its own Points field and can run the Add method.

Trace the state change

new Badge() creates each object with Points = 0 . amber.Add(2) changes amber’s field from 0 to 2; blue remains at 0 because it is a different instance.

Predict, then check the output

The output is 2,0 . A class definition does not create an object. A method call on one object does not automatically change the fields of another.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Different class, different data

What are first.Value and second.Value ?

Method / working:

2. Change the receiver

Continue from Example 1: second.Increase(); What values are stored now? Explain what second identifies.

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Different class, different data CODE: using System; Counter first = new Counter(); Counter second = new Counter(); first.Increase(); first.Increase(); Console.WriteLine(first.Value + ","…

DISCUSS — Compare the sequence of steps or states. Identify the first difference in your methods and justify which step is correct.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Change the receiver Continue from Example 1: second.Increase(); What values are stored now? Explain what second identifies.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.